Swiss Paraplegic Research

Reproducible Reports – R Markdown

Workshop R-Introduction

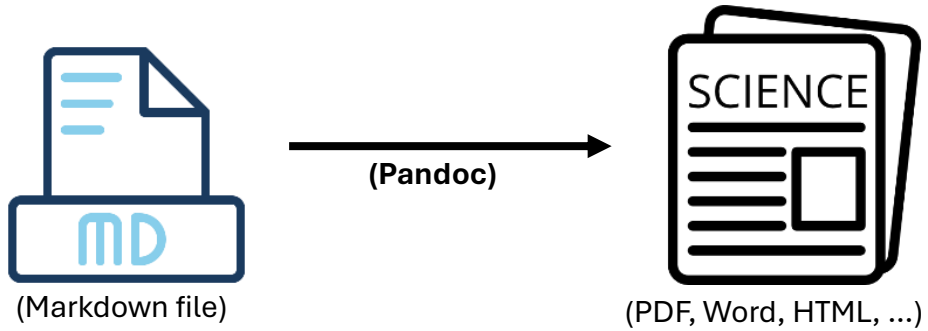
Yannick Rothacher, Data Scientist

19.10.2026



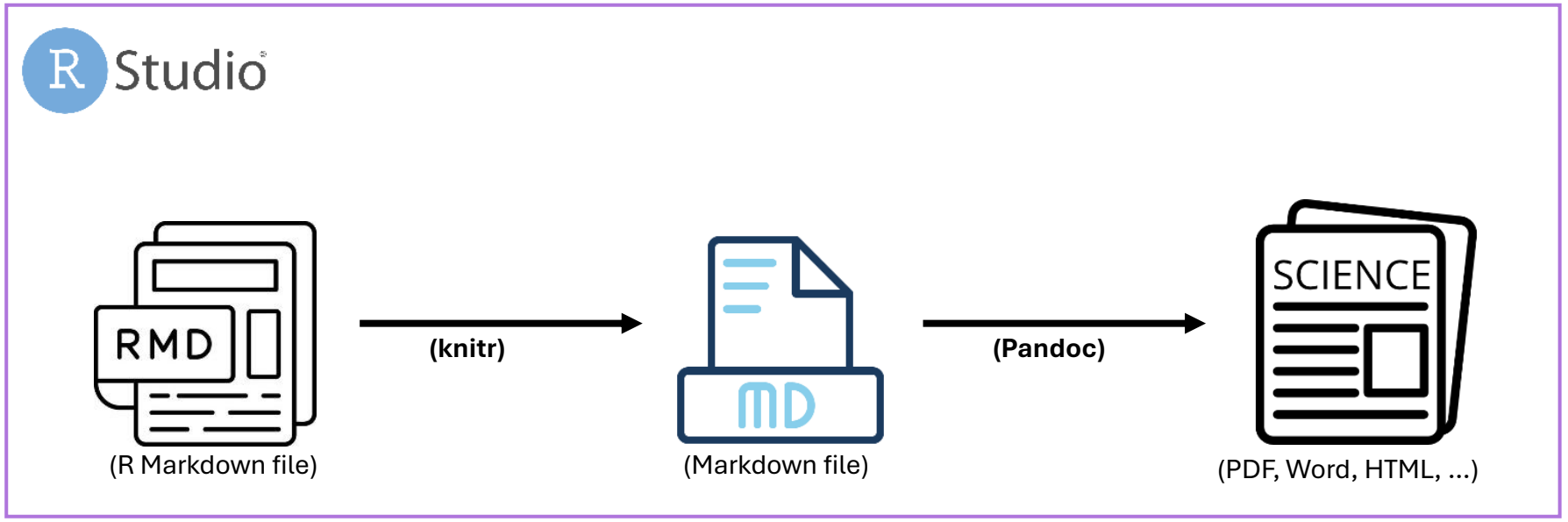
Writing a report in academia

- People are used to writing scientific reports in **Word** (like here at SPF)
  - **However**: There are other options!
- **Markup languages** are an interesting alternative, e.g.:
  - Markdown
  - LaTex
  - …
- With a markup language, we use normal **text** to format a report
- **Markdown** is a simple and popular markup language, let's look at an **example**…



R Markdown (with knitr)

- An **R Markdown** file combines Markdown text with R code
  - The integration of R into Markdown is handled by the **knitr** R-package
- We can create R Markdown files in **RStudio**
- Let's look at an **example**...



R Markdown – Output format

- In RStudio, R Markdown files can be turned into **Word, PDF** or **HTML** files
  - However: To create **PDFs**, a **LaTex** installation is necessary
- The output type is specified in the "**YAML header**"
  - The YAML header is used to specify different document characteristics
- Here, we will focus on using **Word** as output
  - To enable cross-referencing, we will use the **bookdown::word_document2** output

```
---
title: "My Title"
author: "My Name"
date: "`r Sys.Date()`"
output: bookdown::word_document2
lang: en-US
---
```

(YAML header example)

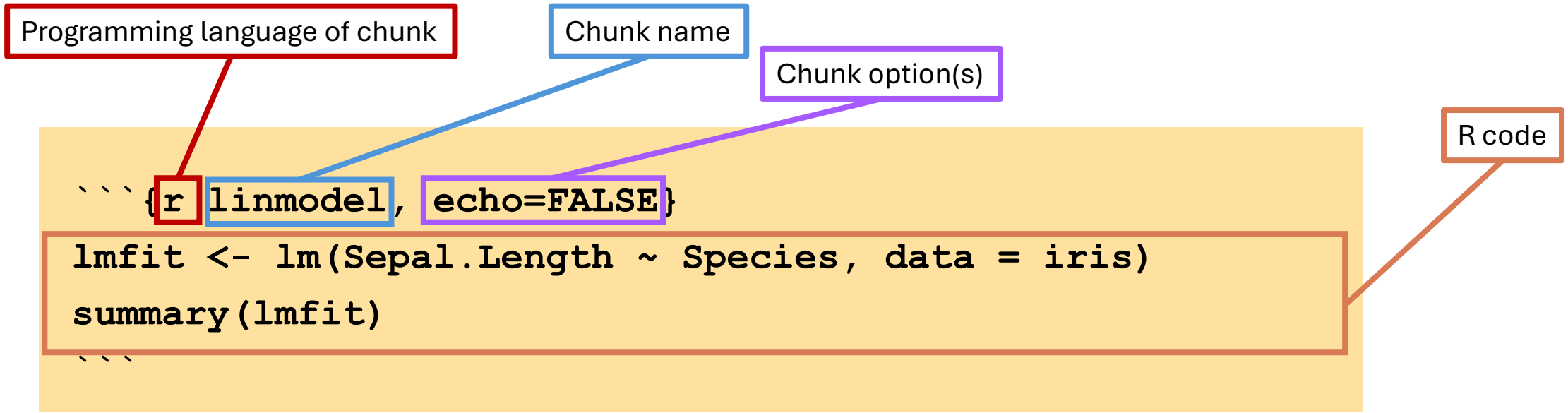
R Markdown – Adding R chunks

- **R chunks** contain R code which is executed when the report is compiled
- The results (and/or code) of an R chunk can be **included in the report**
- Example of an R chunk:

````
```{r linmodel, echo=FALSE}
lmfit <- lm(Sepal.Length ~ Species, data = iris)
summary(lmfit)
```
````

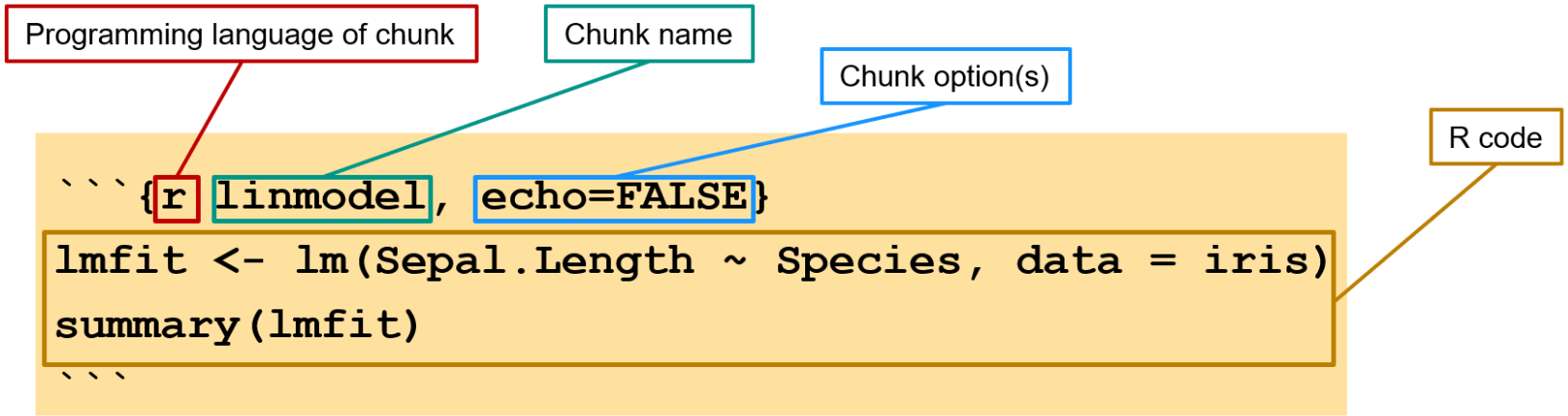
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- Example of an R chunk:



R Markdown – Chunk options

- **Chunk options** can be used to set the behaviour of a chunk
- Some examples:
  - **echo = …** Should R code be shown in document (TRUE or FALSE)
  - **results = …** In what form should results be shown in document (“markup”, “asis”, “hold” or “**hide**”)
  - **include = …** Should anything from the chunk be shown in output (TRUE or FALSE)
  - **eval = …** Should R code be executed when compiling document (TRUE or FALSE)
  - …



R Markdown – Chunk options

- It is often useful to define **chunk option defaults** in a first chunk:

````
```{r setup, include=FALSE}
library(knitr)
opts_chunk$set(echo = FALSE, warnings = FALSE)
```
````

- In the above example…
  - we load needed packages (in this case only **knitr** package)
  - And set chunk option defaults using **opts_chunk$set** function
    - 1) If not specified otherwise, do not show code in report (**echo=FALSE**)
    - 2) If not specified otherwise, do not show potential R-warnings in report (**warnings=FALSE**)

R Markdown – Chunk options

- It is often useful to define **chunk option defaults** in a first chunk:

```
```{r setup, include=FALSE}
library(knitr)
opts_chunk$set(echo = FALSE, warnings = FALSE)
```
```

Include determines whether **any output** should be shown in report (code is still executed)

- In the above example…
  - we load needed packages (in this case only **knitr** package)
  - And set chunk option defaults using **opts_chunk$set** function
    - 1) If not specified otherwise, do not show code in report (**echo=FALSE**)
    - 2) If not specified otherwise, do not show potential R-warnings in report (**warnings=FALSE**)

R Markdown – Inline R code

- It is possible to use R code **inside Markdown text**
- For example, to show a descriptive statistic in a sentence:

```
The average age of the patients was `r mean(dat$age)`
years.
```

- In the report, this will be turned into:

The average age of the patients was 87 years.

R Markdown – Adding tables

- There are different ways **to insert tables** in an R Markdown output
  - Could write it manually in Markdown syntax
  - **Often preferred**: Turn an existing **data frame** into a table
    - We can use the **kable** function for this purpose

```
```{r myTable, echo=FALSE}
kable(dat, caption = "The collected data.")
```
```

- Use the following syntax for **cross-referencing**:

```
As shown in Table \@ref(tab:myTable), the data …
```

  - Will be turned into:

As shown in Table 1.1, the data …

R Markdown – Adding tables

- There are different ways **to insert tables** in an R Markdown output
  - Could write it manually in Markdown syntax
  - **Often preferred**: Turn an existing **data frame** into a table
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This is a function from the **knitr** package:
Need to add **library(knitr)** somewhere above

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```{r myTable, echo=FALSE}
kable(dat, caption = "The collected data.")
```
```

- Use the following syntax for **cross-referencing**:

```
As shown in Table \@ref(tab:myTable), the data …
```

  - Will be turned into:

As shown in Table 1.1, the data …

R Markdown – Adding figures

- Figures can be created and inserted in a **chunk**, for example with:

```
```{r myPlot, out.width = "100%", fig.cap="A simple plot.", echo=FALSE}
plot(x = 1:10, y = 1:10)
```
```

- Use the following syntax for **cross-referencing**:

```
As shown in Figure \@ref(fig:myPlot), the data …
```

  - Will be turned into:

As shown in Figure 1.1, the data …

R Markdown – Adding figures

- Alternatively, one **can create the figure file** manually:

```
```{r, echo=FALSE, results="hide"}
jpeg(filename = 'myPlot.jpeg', width = 10, height = 10,
units = 'cm', pointsize = 10, res = 300)
plot(x = 1:10, y = 1:10)
dev.off()
```
```

- And then **insert the file**:

```
```{r myPlot, out.width = "100%", fig.cap="A simple plot.",
echo=FALSE}
include_graphics(path = "myPlot.jpeg")
```
```

R Markdown – Adding figures

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This is a function from the **knitr** package:
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R Markdown – Adding references

- In R Markdown, the **bibliography and references** are classically managed with a **BibTex file** (.bib)
  - A BibTex file is just a **text file** which contains the **info of the references** in a specific format
  - Let's look at an **example**…

- BibTex files can be **exported** from most reference managers (Zotero, Mendeley, …) or **downloaded** from online data bases (e.g. Google Scholar).

R Markdown – Adding references

- To use a BibTex file…
  - Put the **file** (e.g. references.bib) into the report's folder
  - Add a **bibliography entry** to the YAML header
  - Add a **`# References`** section title at the end of your document (the bibliography is automatically printed at the end of document)

```
---
title: "My Title"
author: "My Name"
date: "`r Sys.Date()`"
output: bookdown::word_document2
lang: en-US
bibliography: references.bib
csl: nature.csl
---
```

- To cite papers, use the following syntax:
  - Citation in brackets: **`[@CitationLabel]`**
  - Multiple citations in brackets: **`[@CitationLabel1; @CitationLabel2]`**
  - Inline citation: **`@CitationLabel`**

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csl: nature.csl
---
```

Can use a **citation style file (.csl)** for different citation styles

- To cite papers, use the following syntax:
  - Citation in brackets: `[@CitationLabel]`
  - Multiple citations in brackets: `[@CitationLabel1; @CitationLabel2]`
  - Inline citation: `@CitationLabel`

Advantages of R Markdown

- Have analysis and text together **in one file**
  - Such reports are called **reproducible** since all analysis steps can be tracked in the R Markdown file

- The **results are generated flexibly**, if something changes in the data the results (including figures) are updated automatically
  - No error-prone **manual copying** of numbers into a Word file

Outlook: Using R and LaTeX (R Sweave files)

- R Markdown is **limited** in its formatting possibilities
- **R Sweave** files combine R chunks with **LaTeX** markup language
  - LaTex offers **extensive control** over document layout and formatting
  - Only output is **PDF**
  - R Sweave files can be generated in RStudio and compiled with **knitr** (or Sweave)

- Let’s look at an example…

Exercise: R Markdown

- Find the exercise at: https://github.com/Swiss-Paraplegic-Research/Workshop-R-Introduction/tree/main/**Part7_RMarkdown/Exercise**